



AI Bootcamp Curriculum (2 Months)

Industry-Aligned, Job-Mapped & Outline (With Weekly Projects)

Theme: "From Foundations to Applications – Learn AI by Doing"

Format: In-person

Methodology: Agile Scrum (Stand-ups, Reviews, Retrospectives)

Delivery Mode: In-Person / Hybrid

Duration: (8 Weeks)

Programme Purpose:

Over 8 weeks, candidates progressively build **one fully functional AI-powered Business Operations Platform**.

Each week adds a new feature, module, or improvement — resulting in a deployable, portfolio-ready system.



Continuous Project Theme:

"AI-Powered Business Operations Assistant"

The final product will be a multi-feature AI system that helps organisations:

- Classify and manage internal requests



- Generate automated responses
- Analyse business data
- Provide insights dashboards
- Assess compliance and risk
- Automate workflows

By Week 8, candidates will have a **fully functional AI-enabled business productivity application**.

Month 1: AI Foundations & Core System Build

What it's about:

This month introduces participants to the core ideas behind Artificial Intelligence. You'll explore foundational AI concepts, real-world applications, key terminology, and ethical considerations. The focus is on **learning the language of AI, understanding how AI impacts industries**, and **experimenting with creative tools like ChatGPT and DALL-E**. You'll also practice teamwork and communication through collaborative research and design projects.

What to expect:

- Exposure to top Coursera courses from IBM, Google, Intel, and DeepLearning.AI
- Hands-on projects such as writing blog posts, designing presentations, and experimenting with generative AI tools
- Group collaboration, storytelling, and ethical discussions
- Foundational knowledge in AI and LLMs to prepare you for deeper applications in

Month 2

Week 1 – Sprint 1: System Design & Ticket Classification Module

Courses Focus:

- **Generative AI: Prompt Engineering Basics**
- **AI For Everyone**
- **Introduction to Artificial Intelligence (AI)**

Sprint Goal: Build the foundation of the platform.

Feature Built: Smart Ticket Classification Engine

Candidates:



- Design the system architecture
- Create categories (HR, IT, Finance, Operations)
- Build a basic AI classifier for incoming requests
- Store classified outputs

Outcome: Functional backend logic for request categorisation.

Mapped Roles: AI Intern, IT Support Assistant, Junior Automation Support

Grading Rubric:

- Technical Skills – Working assistant, correct responses (50%)
- Professional Skills – Usability, explanation, documentation (50%)

Alignment:

- Apply foundational AI concepts to structured tasks

Week 2 – Automated Response Generation Module

Courses Focus:

- **Introduction to Generative AI**
- **AI Essentials**

Sprint Goal: Add intelligent communication capability.

Feature Built: AI Response Generator

Candidates:

- Integrate generative AI
- Build templates for professional responses
- Connect responses to ticket categories
- Implement tone control (formal, friendly, urgent)

Outcome: System now classifies AND responds to requests.

Mapped Roles: Service Desk Analyst, AI Intern, Customer Support Analyst

Grading Rubric:

- Technical Skills – Problem clarity, AI feasibility (40%)
- Professional Skills – Teamwork, presentation, ethics discussion (60%)

Alignment:

- Use AI tools to enhance service delivery



Week 3 – Sprint 2: Data Tracking & Analytics Module

Courses Focus:

- **Generative AI with Large Language Models**
- **AI Foundations: Prompt Engineering with ChatGPT**

Sprint Goal: Introduce data monitoring.

Feature Built: Analytics & Tracking Dashboard

Candidates:

- Store ticket data in structured format
- Analyse trends (volume, categories, response time)
- Build visual dashboard
- Generate weekly summary insights

Outcome: Platform now produces measurable business insights.

Mapped Roles: Junior Data Analyst, Reporting Assistant, AI Intern

Grading Rubric:

- Technical Skills – Effective prompts, useful outputs (60%)
- Professional Skills – Content relevance, clarity, tone (40%)

Alignment:

- Interpret and present structured data

Week 4 – Presentation Sprint: Business Reporting Automation

Courses Focus:

- **Python for Data Science, AI & Development**
- **Supervised Machine Learning: Regression and Classification**

Sprint Goal: Add executive-level reporting.

Feature Built: Automated Weekly Business Report Generator

Candidates:

- Convert dashboard insights into automated summaries
- Generate downloadable PDF or structured report
- Customise reporting per department

Outcome: End-to-end reporting automation integrated.

Mapped Roles: Reporting Analyst, Automation Assistant, Junior Analyst

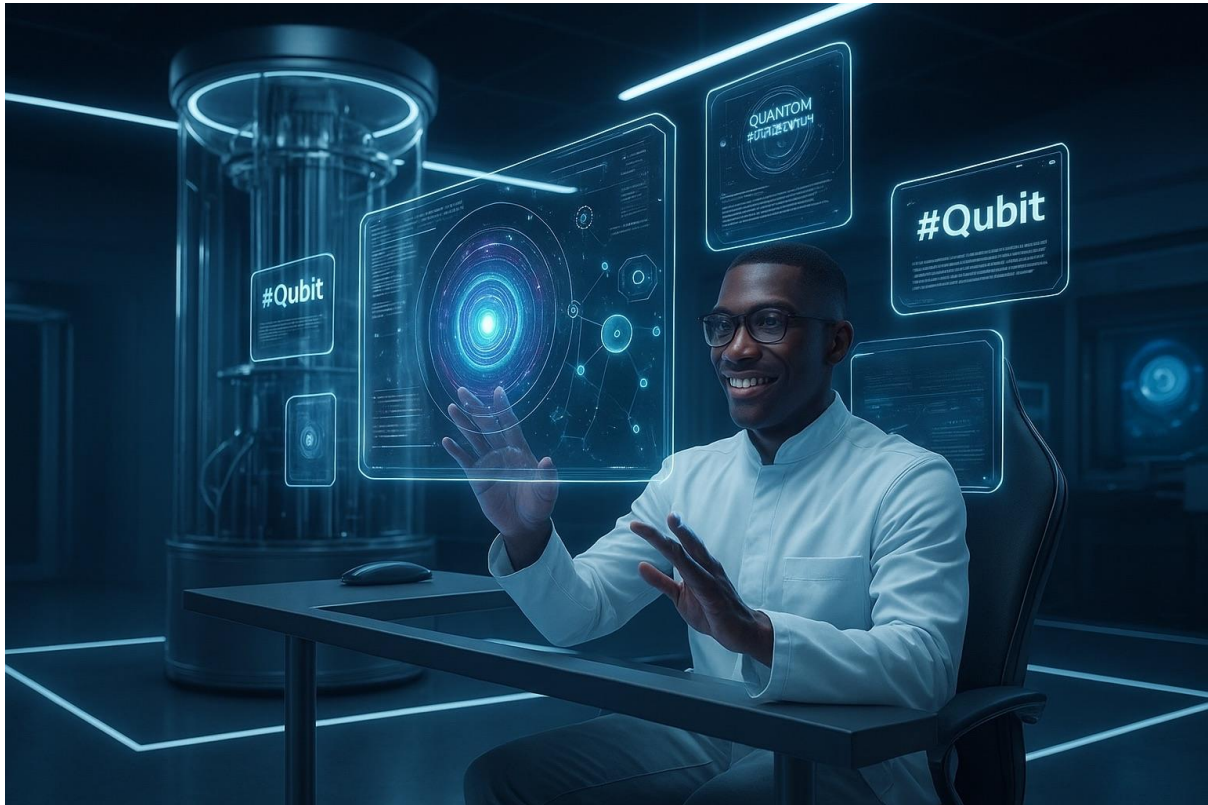


Grading Rubric:

- Technical Skills – Functional tool or prototype (50%)
- Professional Skills – Responsible AI use, presentation quality (50%)

Alignment:

- Translate data into structured business communication



Month 2: Applied AI & Industry Simulation

What it's about:

This month dives into the **practical side of AI development**. You'll explore machine learning workflows, build your first AI model (even without code), and learn how to apply AI responsibly. The capstone project gives you the chance to bring everything together—applying tools, collaborating in a team, and presenting your final work to mentors and potential employers.

What to expect:

- Technical skills development (Python, ML basics, or No-Code tools)
- Building a simple AI model using Google Colab or Teachable Machine
- Group projects exploring ethical AI use and policy



- A final capstone sprint where you'll design and demo your own AI-powered solution
- Career prep: portfolio building, resume workshops, and public showcase

Week 5 – Sprint 3: Predictive Insights Module

Courses Focus:

- **Advanced Learning Algorithms**
- **Unsupervised Learning, Recommenders, Reinforcement Learning**

Sprint Goal: Introduce forecasting capability.

Feature Built: Trend Prediction Component

Candidates:

- **Use historical data**
- **Build simple predictive logic (e.g., ticket surge forecast)**
- **Display projected workload**

Outcome: Platform now supports forward-looking decision-making.

Mapped Roles: Junior Data Analyst, Operations Analyst, AI Intern

Grading Rubric:

- Technical Skills – Accurate analysis, working dashboard (60%)
- Professional Skills – Insight explanation, business relevance (40%)

Alignment:

- Apply analytical models for forecasting

Week 6 – AI Risk & Compliance Layer

Courses Focus:

- **Trustworthy AI: Managing Bias, Ethics, and Accountability**
- **Introduction to Responsible AI**

Sprint Goal: Add governance features.

Feature Built: Compliance & Risk Monitoring Module

Candidates:

- Identify bias risks in response generation



- Implement logging & transparency notes
- Create risk evaluation report feature

Outcome: Platform includes ethical & compliance awareness layer.

Mapped Roles: Compliance Support Officer, AI Governance Assistant

Grading Rubric:

- Technical Skills – Identification of risks and bias (40%)
- Professional Skills – Report clarity, ethical reasoning (60%)

Alignment:

- Evaluate technological systems for ethical risks

Week 7 – Sprint 4: Workflow Automation & System Integration

- **print Goal:** Transform system into automation engine.
- **Feature Built:** End-to-End Workflow Automation
- Learners:
- Automate ticket routing
- Trigger email notifications
- Integrate approval workflows
- Optimise user experience
- **Outcome:** Fully integrated automation system.
- **Mapped Roles:** Junior Automation Specialist, Digital Operations Assistant
- **NQF Alignment:** NQF 6 – Design integrated technology-based solutions
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Grading Rubric:

- Technical Skills – Solution functionality and relevance (60%)
- Professional Skills – Teamwork, documentation, delivery (40%)

Alignment:

- Design integrated technology-based solutions

Week 8 – Deployment, Optimisation & Industry Showcase

Sprint Goal: Finalise and deploy.

Final Deliverables:

- Deployed AI Business Operations Platform
- Technical Documentation
- Business Case Presentation
- Professional Portfolio
- Candidates present to a mock executive or industry panel.

Mapped Roles: AI Intern, Junior Analyst, Automation Specialist, Support Roles

Grading Rubric:

- Technical Skills – Portfolio completeness (40%)
- Professional Skills – Communication, branding, presentation (60%)

Alignment:

- Demonstrate applied competence and workplace readiness

Programme Exit Profile

By the end of this bootcamp, candidates will:

- A multi-feature AI-powered business productivity system
- A deployed, working application
- Technical documentation
- Executive-level presentation skills
- Portfolio-ready industry project

This continuous structure ensures progressive learning while delivering one fully functional, industry-aligned application.